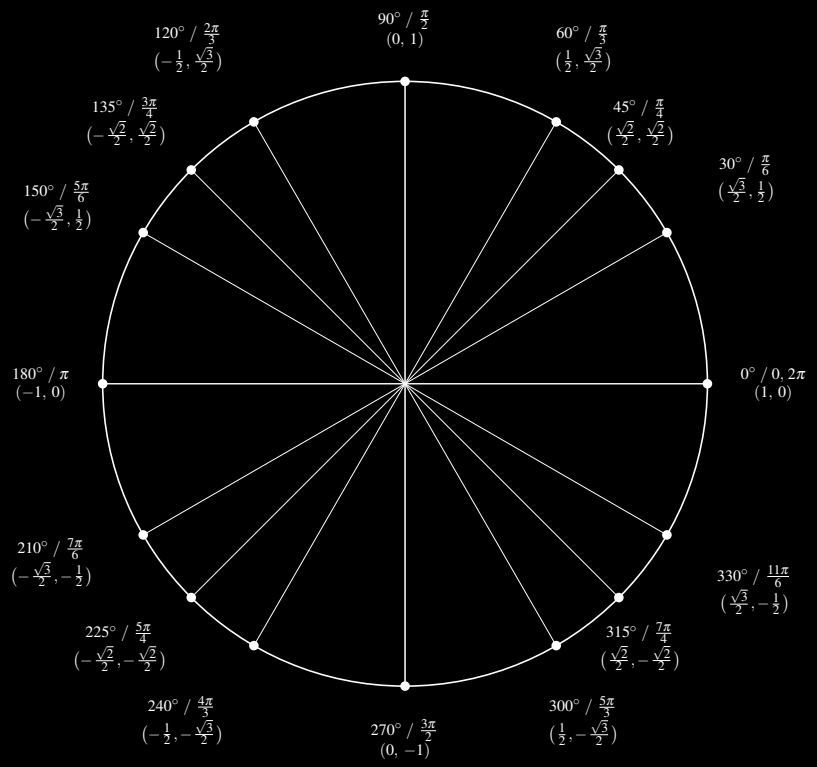
Trigonometry



degrees ↔ radians

- $deg \times \frac{\pi}{180} = \text{rad}$
- $\text{rad} \times \frac{180}{\pi} = deg$

deg	0°	30°	45°	60°	90°
rad	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$

Example: $135° \times \frac{\pi}{180} = \frac{3\pi}{4}$ $\qquad$ $\frac{5\pi}{6} \times \frac{180}{\pi} = 150°$

root-n pattern

$\sin\theta = \frac{\sqrt{n}}{2}$, $\cos\theta = \frac{\sqrt{4-n}}{2}$

θ	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

key identities

pythagorean

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$

Worked: $\sin\theta = \frac{3}{5}$***, θ in QII — find*** $\cos\theta$, $\tan\theta$

1. PYTHAGOREAN
$\cos^2\theta = 1 - \sin^2\theta = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25} = \frac{16}{25}$ $\Rightarrow$
$\cos\theta = \pm\frac{4}{5}$

↓

2. QUADRANT PICKS THE SIGN
QII: $x < 0 \Rightarrow \cos\theta = -\frac{4}{5}$

↓

3. QUOTIENT
$\tan\theta = \frac{3/5}{-4/5} = -\frac{3}{4}$

quotient / reciprocal

- $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{y}{x}$
- $\cot\theta = \frac{\cos\theta}{\sin\theta}$
- $\sec\theta = \frac{1}{\cos\theta}$
- $\csc\theta = \frac{1}{\sin\theta}$
- Unit circle: $(\cos\theta, \sin\theta) = (x, y)$

sum & difference

$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B$
$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}$

Worked: exact $\cos 15°$

1. SPLIT INTO KNOWN ANGLES
$15° = 45° - 30°$

↓

2. APPLY $\cos(A-B)$
$\cos 45°\cos 30° + \sin 45°\sin 30°$

↓

3. EXACT VALUES
$\frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4} = \frac{\sqrt{6}+\sqrt{2}}{4}$

double angle

- $\sin 2A = 2\sin A\cos A$
- $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$
- $\tan 2A = \frac{2\tan A}{1 - \tan^2 A}$

Pick the $\cos 2A$ form matching what you *know*: know $\sin \to 1 - 2\sin^2 A$; know $\cos \to 2\cos^2 A - 1$; both $\to \cos^2 A - \sin^2 A$.

Worked: $\sin 2\theta$, $\cos 2\theta$ ***for*** $\sin\theta = \frac{3}{5}$***, QII***

1. GET THE MISSING PARTNER
from the Pythagorean worked above: $\cos\theta = -\frac{4}{5}$

↓

2. DOUBLE ANGLE
$\sin 2\theta = 2\sin\theta\cos\theta = 2 \cdot \frac{3}{5} \cdot \left(-\frac{4}{5}\right) = -\frac{24}{25}$

↓

3. PICK THE FORM YOU KNOW
know sin:

$$\cos 2\theta = 1 - 2\sin^2\theta = 1 - 2\left(\tfrac{3}{5}\right)^2$$
$$= 1 - 2 \cdot \tfrac{9}{25} = 1 - \tfrac{18}{25} = \tfrac{7}{25}$$

power reduction / half angle

$\sin^2\theta = \frac{1 - \cos 2\theta}{2}$ $\qquad$ $\cos^2\theta = \frac{1 + \cos 2\theta}{2}$

$\sin\frac{A}{2} = \pm\sqrt{\frac{1 - \cos A}{2}}$ $\qquad$ $\cos\frac{A}{2} = \pm\sqrt{\frac{1 + \cos A}{2}}$

The $\pm$ is not optional — the quadrant of $\frac{A}{2}$ (not of A) decides it.

Worked: exact $\sin 22.5°$

1. WRITE AS A HALF
$22.5° = \frac{45°}{2}$: $\quad \pm\sqrt{\frac{1 - \cos 45°}{2}}$

↓

2. RESOLVE THE SIGN
$22.5°$ is QI $\Rightarrow \sin > 0 \Rightarrow$ take $+$

↓

3. SIMPLIFY
$\sqrt{\frac{1 - \sqrt{2}/2}{2}} = \sqrt{\frac{2 - \sqrt{2}}{4}} = \frac{\sqrt{2 - \sqrt{2}}}{2}$

law of sines & cosines

For $\triangle ABC$ with sides a,b,c opposite A,B,C:

sines: $\dfrac{a}{\sin A}=\dfrac{b}{\sin B}=\dfrac{c}{\sin C}$

cosines: $c^2=a^2+b^2-2ab\cos C$; solved for the angle: $\cos C=\dfrac{a^2+b^2-c^2}{2ab}$

Worked: sides $3,5,7$ — the largest angle

1. SOLVE FOR THE ANGLE OPPOSITE $c=7$
$\cos C=\dfrac{9+25-49}{2\cdot3\cdot5}=\dfrac{-15}{30}=-\frac{1}{2}$

↓

2. ARCCOS
$C=\arccos\left(-\frac{1}{2}\right)=120^\circ$ *(negative cosine ⇒ obtuse, automatically)*

SSA caution: law of *sines* with two sides + a non-included angle can give 0, 1, or 2 triangles — always test the supplement $180^\circ-\theta$ as a second candidate.

polar coordinates

$(x,y)\to(r,\theta)$: $r=\sqrt{x^2+y^2}$, $\theta=\arctan\left(\dfrac{y}{x}\right)$

Quadrant check: arctan only returns QI/QIV angles. If $x<0$ (QII/QIII), add 180° (π) to θ.

$(r,\theta)\to(x,y)$: $x=r\cos\theta$, $y=r\sin\theta$

Worked: $(-3,3)\to(r,\theta)$ — quadrant fix, live

1. RADIUS
$r=\sqrt{(-3)^2+3^2}=\sqrt{9+9}=\sqrt{18}=3\sqrt{2}$

↓

2. RAW ARCTAN
$\arctan\left(\frac{3}{-3}\right)=\arctan(-1)=-45^\circ$ *(a QIV angle — but the point sits in QII)*

↓

3. QUADRANT FIX
$x<0$: add 180°: $\theta=135^\circ$ ⇒ $(3\sqrt{2},\ 135^\circ)$

Reverse direction — no traps: $(4,\frac{2\pi}{3})$:
$x=4\cos\frac{2\pi}{3}=-2$, $y=4\sin\frac{2\pi}{3}=2\sqrt{3}$.

Worked: polar equation → rectangular: $r=2\cos\theta$

1. MULTIPLY BY r
$r^2=2r\cos\theta$ *(manufactures the convertible pieces)*

↓

2. SUBSTITUTE
$r^2=x^2+y^2$, $r\cos\theta=x$: $x^2+y^2=2x$

↓

3. COMPLETE THE SQUARE
move the x's together, add $\left(\frac{2}{2}\right)^2=1$ to both sides, factor:
$x^2-2x+y^2=0\ \to\ x^2-2x+1+y^2=1\ \to\ (x-1)^2+y^2=1$
— circle, center $(1,0)$, radius 1

vectors

$\vec{v}=\langle v_1,v_2\rangle$ $\quad|\vec{v}|=\sqrt{{v_1}^2+{v_2}^2}$ $\quad\hat{v}=\dfrac{\vec{v}}{|\vec{v}|}$

$\vec{u}\cdot\vec{v}=u_1v_1+u_2v_2=|\vec{u}||\vec{v}|\cos\theta$

$\text{proj}_{\vec{v}}\vec{u}=\dfrac{\vec{u}\cdot\vec{v}}{|\vec{v}|^2}\vec{v}$ $\quad\text{comp}_{\vec{v}}\vec{u}=\dfrac{\vec{u}\cdot\vec{v}}{|\vec{v}|}$

Worked: $\vec{u}=\langle3,4\rangle$, $\vec{v}=\langle5,0\rangle$

1. DOT & MAGNITUDES
$\vec{u}\cdot\vec{v}=15$, $\quad|\vec{u}|=5$, $\quad|\vec{v}|=5$

↓

2. ANGLE
$\cos\theta=\frac{15}{25}=\frac{3}{5}$ ⇒ $\theta=\arccos(0.6)\approx53.1^\circ$

↓

3. PROJECTION ONTO $\vec{v}$
$\text{comp}_{\vec{v}}\vec{u}=\frac{15}{5}=3$, $\quad\text{proj}_{\vec{v}}\vec{u}=\frac{15}{25}\langle5,0\rangle=\langle3,0\rangle$ *($\vec{v}$ rides the x-axis: the projection is $\vec{u}$'s shadow on it)*

Legend

Color & notes
functions $f,g,\sin$
variables x,y,n
numbers 0–9
operators $+-=$
delimiters $()\,[]$
titles in white
labels · results · pitfalls
$\boxed{u}$ / $\boxed{du}$ substitution

Greek
constants (fixed values):
π ≈3.14159
γ fluid wt density
ρ weight per length
variable & operators:
θ angle (varies)
Δ "change in"
Σ sum of terms
notation, not values:
$\Theta\,\Omega\,\omega$ growth rate, like O

Calculus & relations
$\int$ integral
$\sum$ sum
lim limit
∞ infinity
$\to$ approaches
$\Rightarrow$ implies
$\Leftrightarrow$ iff
$\sqrt{\ }$ root
$\approx$ approx
$\neq<\leq\geq$ compare
$\pm\mp$ plus / minus
$\cdot\times\div$ mult / div
$\circ$ compose

Vectors
$\langle\rangle$ components
$\|\vec{v}\|$ magnitude
$\vec{v}$, $\hat{v}$ vector, unit
$\mathbf{ijk}$ basis
$\vec{u}\cdot\vec{v}$ dot product
proj projection

Logic & sets
$\forall$ for all
$\exists$ exists
$\neg$ not
$\wedge\vee$ and / or
$\oplus$ xor / sym. diff
$\in\notin$ in / not in
$\subseteq\subset$ subset
$\cup\cap$ union / inter.
$\setminus$ set minus
$\equiv$ equiv / congruent
$|$ divides / $|x|$ / s.t.
$\lceil\ \rceil$ ceiling
$\binom{n}{k}$ choose
deg degree
mod modulo